

Response to the Editor and Reviewers

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Ehsan Roohi

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Dear Professor Bucci and Reviewers,

Thank you for the careful and constructive review of “Physics-Guided Neural Surrogates for Canonical Compressible Thermal-Fluid Relations.” The manuscript has been substantially restructured. New and revised manuscript text is shown in blue; a clean version is also supplied. All numerical claims now come from a public, one-command repository with unit tests, machine-readable CSV evidence, and vector figures. We have removed qualitative overclaims and explicitly report negative results, classical baselines, coverage, uncertainty, and limitations.

Summary of major changes

1. Reframed the work as a controlled SciML benchmark for multi-valued manifolds, singular limits, and constraint-aware representations.
2. Corrected the Rayleigh inverse to use T_0/T_0^* for a sonic branch split; static T/T^* peaks at $M = 1/\sqrt{\gamma}$ and was unsuitable for the previous split.
3. Replaced the raw Fanno output with an exact quadratic, positive sonic transform and added a controlled raw-versus-structured ablation.
4. Added weak/strong oblique inverse experts and a naive-MLP failure experiment; replaced soft endpoint anchors with a hard analytical output envelope.
5. Added bounded nozzle and shock-tube outputs, a two-input shock-tube model, robust Brent reference roots, and residual checks.
6. Added blind L_2/L_∞ /MAE/RMSE metrics, near-limit bins, three-seed ensembles, sample-size sensitivity, interpolation baselines, and measured CPU timing.
7. Added a limitations/failure-modes section and replaced all claims such as “infinite precision” or “near-perfect” with numerical statements.

Reviewer 1

Reviewer comment. *The novelty relative to existing PINNs and surrogate models is not sufficiently clear.*

Response. We rewrote the Introduction and the new section “What the network learns and what it does not.” We no longer imply a new general-purpose neural architecture or a strict PDE-residual PINN. The novelty claimed is a reproducible benchmark and controlled evidence for branch experts, singular-limit features, bounded output transforms, and hybrid analytical reconstruction. We explicitly position the work as complementary to PINNs and neural operators.

Reviewer comment. *Provide quantitative L_2 and L_∞ errors for all five problems.*

Response. Table 2 now reports relative L_2 , relative L_∞ , MAE, and maximum absolute error for the blind inverse task of every problem. The relative L_2 values are 1.642×10^{-3} (Rayleigh), 3.431×10^{-3} (Fanno), 7.990×10^{-4} (oblique), 1.936×10^{-3} (nozzle), and 9.490×10^{-4} (shock tube). The exact CSV is `results/revision/primary_metrics.csv`.

Reviewer comment. *Explain and validate the branch-decomposition implementation.*

Response. Section 2 and Table 1 now state the branch variables and bounds. Rayleigh and Fanno use independently trained subsonic and supersonic experts. The oblique inverse has distinct weak/strong outputs bounded between the Mach angle and $\pi/2$. Figure 2 validates the need for this design: a naive oblique MLP has branch-distance MAE 0.4246 rad, whereas the branch-aware model has MAE 5.076×10^{-4} rad and 100% correct ordering.

Reviewer comment. *Report the complete architecture, hyperparameters, and training procedure.*

Response. Section 3.1 now gives software versions, scaling, activation, layer widths, L-BFGS settings, regularization, tolerances, maximum iterations/function evaluations, sample counts, blind-grid sizes, and seeds. These settings are also executable in `src/gasdynbench/modeling.py` and `run_revision.py`.

Reviewer comment. *Compare against classical methods and existing machine-learning approaches.*

Response. Section 4.3 compares branch-wise PCHIP/piecewise-linear interpolation, bracketed Brent roots, a naive MLP, and the physics-guided MLP. We report both accuracy and coverage. Interpolation is faster and more accurate for covered one-dimensional tables and is preferred there. New Table 5 extends the same shock-tube relation from two to five physical inputs under a matched offline-state budget. The five-dimensional interpolation and MLP relative L_2 errors are 4.885% and 0.1771%, respectively, while interpolation remains slightly faster. This confines the neural advantage to compact constrained approximation as dimension increases, not universal numerical superiority.

Reviewer comment. *Demonstrate actual efficiency relative to root finding or interpolation.*

Response. We removed the former “indicative” timing table. Table 4 and Figure 4 now report warm-up, seven-repeat median CPU timings with IQR stored in `timing.csv`. For 5000 queries the MLP is 96–1057 times faster than repeated bracketed roots, but interpolation is 4–11 times faster than the MLP. The added dimensional audit likewise reports offline storage and online timing; offline training cost is excluded from inference speed-up and remains explicit.

Reviewer comment. *Assess robustness near singularities and discontinuities.*

Response. Section 4.2 and Figure 3 add local bins near Rayleigh/Fanno sonic states, oblique detachment, nozzle throat/exit limits, and high shock-tube pressure ratio. The Fanno transform enforces the quadratic sonic limit. Shock-tube discontinuities are reconstructed analytically rather than learned as a smooth field, and equation residuals are reported.

Reviewer comment. *Test generalization to unseen parameter ranges.*

Response. We added a dedicated edge-holdout experiment in new Table 3 and its machine-readable CSV. Each model is trained only on an interior box and tested on omitted low/high parameter bands inside the declared global physical domain. Edge-band relative L_2 errors are 0.0341 (Rayleigh), 0.0430 (Fanno), 0.00755 (oblique), 0.0106 (nozzle), and 0.00432 (shock tube), with 100% hard-bound validity. The increased Rayleigh/Fanno errors are discussed explicitly. We call this controlled short-range extrapolation and continue to reject use beyond the global domain.

Reviewer comment. *Quantify thermodynamic consistency.*

Response. Section 4.2 now reports entropy-value L_∞ discrepancies of 0.0205 (Rayleigh) and 0.0175 (Fanno), and zero entropy-maximum location error on the blind grid. We also report non-negativity, branch-ordering, nozzle-location, pressure-bound, and equation-residual diagnostics in `physical_diagnostics.csv`. We no longer claim exact thermodynamic consistency for unconstrained property outputs.

Reviewer comment. *Examine sensitivity to input selection and training data size.*

Response. Figure 2 isolates formulation sensitivity: raw versus quadratic-positive Fanno output and naive versus branch-aware oblique inputs/outputs. Figure 3 reports 200/500/900-sample studies as mean $\pm$ SD over three seeds. The revised text discusses the observed plateau and Fanno optimizer sensitivity rather than assuming monotonic improvement.

Reviewer comment. *Discuss inverse formulation stability.*

Response. Section 2 states the branch-specific coordinates, bounded reconstruction, and sonic-distance features. Near-limit L_∞ errors are reported separately. The revised code uses bracketed roots for reference data and never relies on an unconstrained `fsolve` initial guess.

Reviewer comment. *Add uncertainty quantification or confidence estimates.*

Response. Section 3.2 and Section 4.2 add three-seed ensembles and report mean $\pm$ SD error norms. We label this correctly as training-seed variability rather than calibrated probabilistic UQ. The Limitations section identifies calibrated uncertainty as future work and warns that small ensemble spread does not justify out-of-domain use.

Reviewer comment. *State limitations and failure cases.*

Response. A dedicated six-point Limitations section has been added. It covers perfect-gas/one-dimensional assumptions, finite domains, uncalibrated ensemble uncertainty, partial rather than complete physical enforcement, cases where interpolation dominates, and omission of offline cost. Figure 2 preserves two negative results.

Reviewer comment. *Improve figure labels, concision, and grammar.*

Response. All result figures were regenerated as vector PDFs and 400-dpi PNGs with consistent DejaVu Serif typography, mathematical labels, colorblind-safe colors, and explicit captions. The manuscript was condensed and line-edited; typographical errors such as “he Throat Limit” and ambiguous axis descriptions were removed.

Reviewer 2

Reviewer comment. *Frame the work as a foundational SciML benchmark for multivalued manifolds, singular limits, and generalizable strategies.*

Response. This is now the central framing in the title-page abstract, Introduction, and Conclusions. The manuscript distinguishes the benchmark contribution from claims of a universal architecture.

Reviewer comment. *Show the failure of a naive MLP for the multivalued inverse.*

Response. Figure 2 includes the requested controlled experiment. Duplicated $(M, \theta/\theta_{\max})$ inputs with weak and strong labels cause the naive model to predict between branches. Quantitative branch-distance MAE falls from 0.4246 rad to 5.076×10^{-4} rad with branch experts.

Reviewer comment. *Use a weighted loss or provide granular near-singular error.*

Response. We chose the second option and added four local distance bins for every problem in `near_singular_metrics.csv` and Figure 3. The Fanno and oblique output transforms additionally impose exact limiting structure, which is more transparent than a tuned loss weight.

Reviewer comment. *Discuss whether the model learns a representation or merely an interpolator.*

Response. A new dedicated section addresses this question. We conclude that the MLP is a parametric approximant of exact data; the useful learned representation is its branch/bound structure. PCHIP is superior for a fixed one-dimensional table. A concise matched-budget audit now identifies the higher-dimensional regime in which grid resolution is lost while keeping this qualification explicit.

Reviewer comment. *Study scaling with the amount of training data.*

Response. Figure 3 now reports 200, 500, and 900 samples over three seeds. We discuss the observed plateau and non-monotonic Fanno optimization variability rather than hiding it. All rows are available in `training_size_scaling.csv`.

Reviewer comment. *Clarify whether boundary anchors are hard or soft.*

Response. The earlier samples were soft anchors because they only entered the supervised data. We corrected this terminology. The revised direct oblique model uses the hard output transform $\hat{\theta} = q(1 - q) \exp h_{\theta}$, which is exactly zero at the Mach-wave and normal-shock endpoints.

Reviewer comment. *Add recent relevant studies, including Rehman et al., Case Studies in Thermal Engineering 37 (2022) 102221.*

Response. We added and discussed K.U. Rehman, W. Shatanawi, and A.B. Çolak, “Thermal analysis of flowing stream in partially heated double forward-facing step by using artificial neural network,” *Case Studies in Thermal Engineering* 37 (2022) 102221, DOI: 10.1016/j.csite.2022.102221. The Introduction explains its relevance as a thermal-fluid MLP precedent and distinguishes our focus on branch/singularity benchmarks and controlled ablations.

Repository and reproducibility

The revised public project is <https://github.com/Ehsan-Roohi/GasDynamicsSciML>. It contains the package, analytical tests, CI workflow, configuration, exact CSV evidence, vector figures, highlighted and clean manuscript sources, and this response letter. No qualitative value is inserted into the manuscript without a corresponding generated table.

We appreciate the reviewers’ recommendations, which materially improved the correctness, positioning, and reproducibility of the study.

Sincerely,

Ehsan Roohi